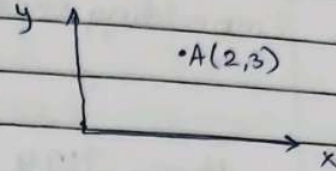


②

Date: / /

Vector maths

- A vector is an ordered set of numbers:
 $A = [2, 3]$



- The vector tells us the point's position relative to the origin.

- It can also represent a direction and magnitude.

- Hence, A vector is a collection of nos. that can represent a point/location or a direction in a mathematical space.

Suppose, $\text{dog} = [0.8, 0.7]$

$\text{puppy} = [0.9, 0.8]$

and $\text{car} = [-0.3, 0.9]$

The machine needs to answer which vector is most similar to dog?

The machine requires mathematical operations that let us compare them.

The important ones are →

i) **Dot product** → measures how strongly 2 vectors point in the same direction

ii) **Magnitude** → measures the length of a vector.

iii) **Cosine similarity** → measures the angle b/w vectors.

iv) **Distance** → measures how far apart vectors are.

$$\text{if } A = (x, y),$$

$$|A| = \sqrt{x^2 + y^2}$$

Date: / /

i) Dot product $\Rightarrow$

$$\text{suppose, } A = [2, 3], B = [4, 5]$$

$$\text{then, } A \cdot B = (2 \times 4) + (3 \times 5) = 23$$

case 1) Vectors are in same direction

Dot product tends to be large & positive.

case 2) Vectors are in opp. direction

Dot product becomes small or negative.

case 3) Vectors are perpendicular

Dot product is zero.

$$\text{Mathematically: } A \cdot B = |A| |B| \cos \theta$$

length of A

length of B

$\theta \rightarrow$ angle b/w them.

Therefore, dot product contains info about both vector length and directional alignment.

Problem with raw dot product:-

$$\text{suppose! } A = [1, 0], B = [1, 0] \Rightarrow A \cdot B = 1$$

$$\text{now, multiply B by 100} \rightarrow C = [100, 0]$$

$$A \cdot C = 100$$

here, dot product says 100 instead of 1, though direction of original vectors remain same.

Does this mean C is more aligned? $\Rightarrow$ not necessarily.

Vector C is just much longer.

This is why we normalize by vector magnitude.

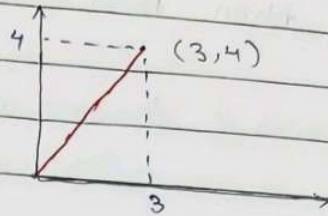
Date: / /

ii) Vector magnitude

• Magnitude simply means the length of a vector.

for, $A = [3, 4]$

$$\begin{aligned} |A| &= \sqrt{3^2 + 4^2} \rightarrow \text{Pythagorean thm} \\ &= \sqrt{25} \Rightarrow \\ &= 5 \end{aligned}$$



for, vector with d-dimensions

$$A = [a_1, a_2, a_3, \dots, a_d]$$

the magnitude is,

$$|A| = \sqrt{\sum_{i=1}^d a_i^2}$$

so, we are simply calculating the length of an arrow in potentially thousands of dimensions!

iii) Cosine similarity

$$\cos \theta = \frac{A \cdot B}{|A| |B|}$$

This tells how similar are the directions of these 2 vectors, independent of their absolute length.

iv) Distance $\Rightarrow$ How far apart are these 2 vectors?

suppose, $A = (2, 3)$, $B = (5, 7)$.

~~first~~ so by euclidean distance formula,

$$\begin{aligned} \text{distance}(A, B) &= \sqrt{(5-2)^2 + (7-3)^2} \\ &= \sqrt{9+16} \\ &= \boxed{5} \end{aligned}$$

Why is this useful for embedding?

suppose, dog = $[0.8, 0.7]$

cat = $[-0.2, 0.1]$

puppy = $[0.9, 0.87]$

conceptually.

If dog & puppy are close in the learned embedding space, their euclidean distance could be small.

so distance can be used as similarity proxy:

smaller distance $\rightarrow$ more similar

Cosine similarity $\Rightarrow$ checks if vectors point in a similar direction

Euclidean distance $\Rightarrow$ How far are these points?

3# What an embedding actually represents

Imagine creating a map of a city.

Places with similar functions tend to appear in similar regions. (Restaurants, hotels, etc.)

Now, imagine a ML model doing something analogous, except instead of mapping physical locations, it learns to map concepts.

The model is learning a representation space in which relationships b/w concepts become mathematically useful.

This is the idea behind embeddings.

